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DR SONG ACADEMY

NCEA Level 3

Calculus

AS 91579

Integration

Learning Workbook

2026 Edition

Achievement Standard 91579

Apply Integration Methods in Solving Problems

Achievement Criteria

Achievement	Achievement with Merit	Achievement with Excellence
Apply integration methods in solving problems.	Apply integration methods, using relational thinking, in solving problems.	Apply integration methods, using extended abstract thinking, in solving problems.

Notes

1. Apply integration methods in solving problems involves:
   - selecting and using methods
   - demonstrating knowledge of concepts and terms
   - communicating using appropriate representations.

   Relational thinking involves one or more of:
   - selecting and carrying out a logical sequence of steps
   - connecting different concepts or representations
   - demonstrating understanding of concepts
   - forming and using a model;

   and also relating findings to a context, or communicating thinking using appropriate mathematical statements.

   Extended abstract thinking involves one or more of:
   - devising a strategy to investigate or solve a problem
   - identifying relevant concepts in context
   - developing a chain of logical reasoning, or proof
   - forming a generalisation;

   and also using correct mathematical statements, or communicating mathematical insight.

2. *Problems* are situations that provide opportunities to apply knowledge or understanding of mathematical concepts and methods. Situations will be set in real-life or mathematical contexts.

3. Methods are selected from those related to:
   - integrating power, polynomial, exponential (base e only), trigonometric, and rational functions
   - reverse chain rule, trigonometric formulae
   - rates of change problems
   - areas under or between graphs of functions, by integration
   - finding areas using numerical methods, eg the rectangle or trapezium rule
   - differential equations of the forms $y' = f(x)$ or $y'' = f(x)$ for the above functions or situations where the variables are separable (eg $y' = ky$) in applications such as growth and decay, inflation, Newton's Law of Cooling and similar situations.

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Chapter 1

Fundamental Integration

LEARNING OBJECTIVES

By the end of this chapter, you should be able to:

1. **Understand the Integral Sign:** Interpret integration as the mathematical process of finding antiderivatives.
2. **Apply the Power Rule:** Integrate polynomial expressions seamlessly.
3. **Perform Algebraic Pre-conversion:** Transform surds and rational fractions into index form to enable integration.
4. **Integrate Transcendental Functions:** Apply the foundational formulae for exponential (e^{kx}) and logarithmic ($\ln|x|$) functions.
5. **Integrate Trigonometric Functions:** Recognise and integrate standard trigonometric functions natively provided on the NCEA formula sheet.

Integration is fundamentally the reverse process of differentiation. If differentiating a function $F(x)$ produces the gradient function $f(x)$, then integrating $f(x)$ returns us to the original function $F(x)$.

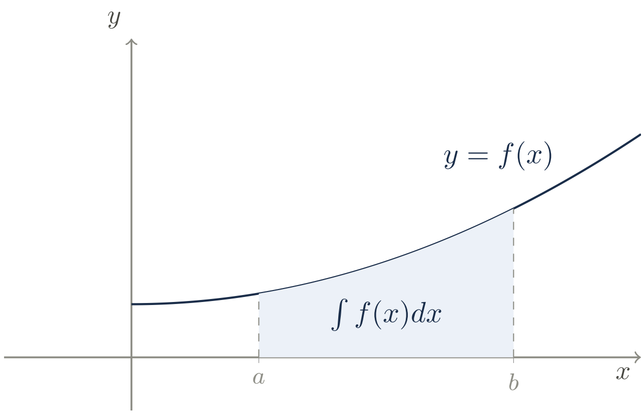
This process is formally called **antidifferentiation.**

$$\int f(x)\,dx \quad = \quad F(x) \quad + \quad c$$

KEY IDEA

The Anatomy of an Integral:

- $\int$: The integral sign (an elongated 'S', representing a continuous sum).
- $f(x)$: The **integrand**, the function being integrated.
- dx : The differential, indicating that the integration is performed with respect to the variable x.
- $+c$: The **constant of integration**. Since the derivative of any constant is zero, there are infinitely many original functions that could produce $f(x)$. The $+c$ acknowledges this family of parallel curves.



COMMON ERROR

The Missing Constant

In NCEA examinations, failing to include the constant of integration ($+c$) when evaluating an indefinite integral is a severe technical error that will immediately invalidate your solution. Always append $+c$ as your final step!

To integrate polynomial terms of the form ax^n, we reverse the differentiation power rule. Instead of multiplying by the power and subtracting one, we **add one to the power** and then **divide by the new power**.

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + c \quad \text{for } n \neq -1$$

EXAM TIP

Constant Terms

The integral of a standalone constant k is simply $kx + c$. (e.g., $\int 5 \, dx = 5x + c$). This is because differentiating $5x$ returns 5.

Example 1

(NCEA 2015) Find

$$\int \left(3 - \frac{5}{x^2}\right) dx$$

Solution:

1. **Prepare the expression:** Rewrite the fraction using a negative index.

$$\int (3 - 5x^{-2}) \, dx$$

2. **Integrate term by term:**
   - The integral of 3 is $3x$.
   - For $-5x^{-2}$, add 1 to the power ($-2 + 1 = -1$), then divide by -1.

$$\frac{-5x^{-1}}{-1} = +5x^{-1}$$

3. **Final Answer:**

$$3x + 5x^{-1} + c \quad \text{or} \quad 3x + \frac{5}{x} + c$$

Example 2

(NCEA 2018) Find

$$\int \left(6x - \frac{8}{x^3}\right) dx$$

Solution:

1. **Prepare the expression:** Rewrite the fraction.

$$\int (6x - 8x^{-3})\, dx$$

2. **Integrate:** Apply the power rule.

$$\frac{6x^2}{2} - \frac{8x^{-2}}{-2} + c$$

3. **Simplify:**

$$3x^2 + 4x^{-2} + c \quad \text{or} \quad 3x^2 + \frac{4}{x^2} + c$$